Bayesian Volatility and Tail Risk Modeling for Financial Returns Using MCMC Methods

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1 Introduction

Financial markets exhibit time-varying risk. Periods of calm are often followed by periods of heightened volatility, especially during economic downturns. Modeling this volatility is crucial for effective risk management. Classical volatility models such as the Generalized Autoregressive Conditional Heteroskedasticity (GARCH) model are widely used, but they rely on point estimates and often assume Gaussian innovations. These assumptions can lead to an underestimation of risk, particularly in the tails of the return distribution.
The main research question of this project is:

> *How does Bayesian modeling of volatility improve uncertainty quantification and tail risk estimation compared to classical volatility approaches such as GARCH when parameters are estimated via MCMC?*

The project uses financial market data from a major equity index or exchange-traded fund (ETF), such as the S&P 500 (SPY). Daily adjusted closing prices over the last 10–20 years will be collected. This period includes both calm markets and highly volatile episodes, such as the 2008 financial crisis and the COVID-19 market crash.
From the price data, log returns are computed as:

$$r_t = \log\left(\frac{S_t}{S_{t-1}}\right),$$

where S_t denotes the adjusted closing price on day t.

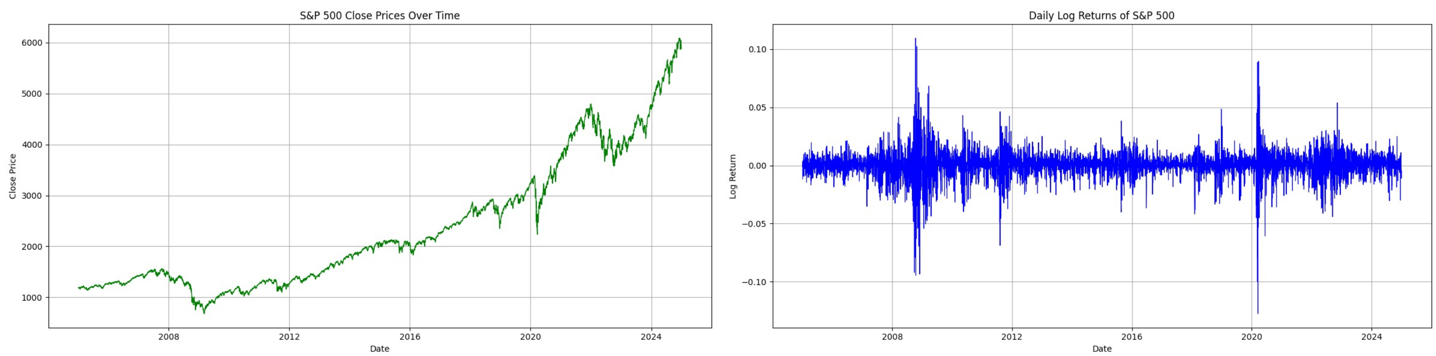


Figure 1: S&P 500 close prices and log returns over January 1st 2005 January 1st 2025

2 Methods

This project applies Bayesian statistical modeling and Markov Chain Monte Carlo (MCMC) inference to financial time series. Two main models are implemented and compared.

2.1 Bayesian GARCH(1,1)

The first model is a Bayesian Generalized Autoregressive Conditional Heteroskedasticity (GARCH) model, which is widely used in financial econometrics. Rather than estimating parameters via maximum likelihood, a Bayesian framework is adopted by placing prior distributions on the model parameters.
The Bayesian GARCH(1,1) model is defined as:

$$r_t = \sigma_t \varepsilon_t, \quad \varepsilon_t \sim \mathcal{N}(0,1),$$
$$\sigma_t^2 = \omega + \alpha r_{t-1}^2 + \beta \sigma_{t-1}^2.$$

The parameters estimated are:

- ω — long-run volatility level,
- α — reaction of volatility to past returns,
- β — persistence of volatility.

2.1.1 Parameter Estimation

Classical estimation methods typically use likelihood estimation, by the model assumptions:

$$r_t = \sigma_t \varepsilon_t, \quad \varepsilon_t \sim \mathcal{N}(0,1)$$

The likelihood function is then given by:

$$\mathcal{L}(r_1, \dots r_n \mid \theta) = \prod_{t=1}^{n} \frac{1}{\sqrt{2\pi\sigma_t^2}} \exp\left(-\frac{r_t^2}{2\sigma_t^2}\right)$$

and $\hat{\theta}_n = (\hat{\omega}_n, \hat{\alpha}_n, \hat{\beta}_n)$ which minimizes this function. Following the following constraint:

$$\alpha + \beta < 1 \quad \text{and} \quad \alpha, \beta, \omega > 0$$

which guaranties a stationary solution with a finite expected value.
For this project we use the Metropolis–Hastings algorithm and follow the adaptive proposal density construction approach proposed in [1]. Specifically we are using a 3-dimensional multivariate student's t-distribution. given by:

$$g(\theta) = \frac{\Gamma((\nu+3)/2)/\Gamma(\nu/2)}{\det \Sigma^{1/2}(\nu\pi)^{3/2}} \left[1 + \frac{(\theta - M)^t \Sigma^{-1}(\theta - M)}{\nu}\right]^{-(\nu+3)/2}$$

Where θ and M vectors are given by:

$$\theta = \begin{bmatrix} u_1 \\ u_2 \\ u_3 \end{bmatrix}, M = \begin{bmatrix} M_1 \\ M_2 \\ M_3 \end{bmatrix} \quad \text{and} \quad M_i = \mathbb{E}[\theta_i]$$

Σ is the covariance matrix. The multivariate Student-t proposal is chosen to better accommodate the heavy-tailed geometry of the posterior distribution. We chose $\nu = 10$ because it provides moderately heavy tails while close to Gaussian.
Due to our constraint problem we will map our problem space to an unconstrained problem space defined by:

$$u = (u_1, u_2, u_3) \in \mathbb{R}^3$$

where:

$$\begin{aligned} & u_1 = \log \omega \quad \text{then} \quad \omega = e^{u_1} \implies \omega > 0 \\ u_2 = log\left(\frac{\rho}{1-\rho}\right), \quad & \rho = \alpha + \beta \quad \text{then} \quad \rho = \sigma(u_2) = \frac{1}{1+e^{-u_2}} \implies \alpha + \beta \in (0, 1) \\ u_3 = log\left(\frac{\delta}{1-\delta}\right), \quad & \delta = \frac{\alpha}{\alpha + \beta} \quad \text{then} \quad \alpha = \rho\delta, \quad \beta = \rho(1-\delta) \implies \alpha, \beta > 0 \end{aligned}$$

For each parameter (ω, ρ, δ) we will place some non-informative priors. Use:

$$\begin{aligned} \omega &\sim \text{Gamma}(2, 2 \cdot 10^5) \\ \rho &\sim \text{Beta}(2, 2) \\ \delta &\sim \text{Beta}(2, 2) \end{aligned}$$

The prior for ω reflects the scale of the observed returns. We, assume independence between all parameters and derive the full log posterior:

$$\log p(\omega, \rho, \delta \mid r_1, \dots, r_n) = \log \mathcal{L}(r_1, \dots r_n \mid \omega, \rho, \delta) + \log p(\omega, \rho, \delta) + \log \left|\frac{\partial(\omega, \rho, \delta)}{\partial(u_1, u_2, u_3)}\right|$$

Where:

$$\begin{aligned} \log \mathcal{L}(r_1, \dots, r_n \mid \omega, \rho, \delta) &= -\frac{1}{2}\sum_{t=1}^{n}\left(\log(2\pi) + \log \sigma_t^2 + \frac{r_t^2}{\sigma_t^2}\right). \\ \log \left|\frac{\partial(\omega, \rho, \delta)}{\partial(u_1, u_2, u_3)}\right| &= \log \omega + \log(\rho(1-\rho)) + \log(\delta(1-\delta)) \\ \log p(\mu, \phi, \sigma^2, \nu) &= \log p(\mu) + \log p(\phi) + \log p(\sigma^2) + \log p(\nu) \end{aligned}$$

2.1.2 Algorithm

The proposal density is a multivariate Student-t distribution

$$u^* \sim t_\nu(M, \Sigma),$$

where M and Σ denote the empirical mean and covariance of previously sampled values in the transformed parameter space. To construct an adaptive proposal, we first run a short pilot Metropolis simulation to obtain preliminary values. These values are used to estimate M and Σ. During the main sampling phase, M and Σ are updated using accumulated samples.

2.1.3 Results

The results of the simulation on our S&P 500 data are as follows:

Table 1: Posterior summaries for Bayesian GARCH(1,1) parameters

Parameter	Mean	Median	2.5%	97.5%
ω	2.84×10^{-6}	2.82×10^{-6}	2.21×10^{-6}	3.55×10^{-6}
α	0.1294	0.1291	0.1103	0.1506
β	0.8455	0.8458	0.8229	0.8663
$\rho = \alpha + \beta$	0.9750	0.9751	0.9655	0.9837
$\delta = \alpha/(\alpha + \beta)$	0.1328	0.1325	0.1130	0.1544

The posterior mean of $\alpha \approx 0.13$ indicates a moderate short-run reaction of volatility to new return shocks, implying that large returns generate noticeable increases in conditional variance. The posterior mean of $\beta \approx 0.85$ reflects strong volatility persistence, indicating that shocks to volatility decay slowly over time. The posterior mean of $\omega \approx 2.8 \times 10^{-6}$ determines the long-run variance level of the process and is consistent with the empirical scale of daily return variability.

The following is a graph of 5 simulated volatility paths against the real volatility path of our data:

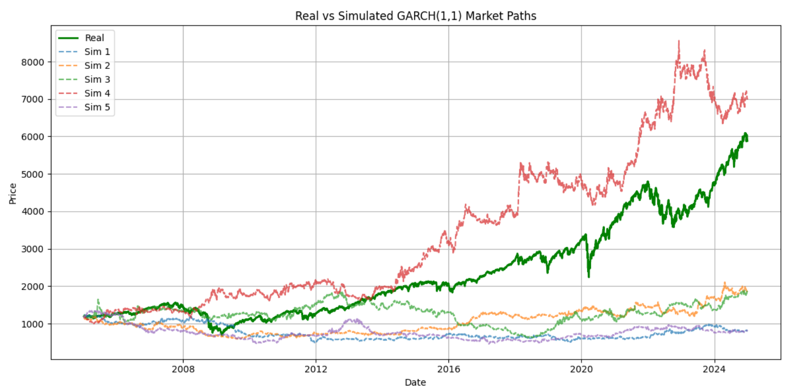


Figure 2: Real vs Simulated GARCH(1,1) volatility paths

2.2 Bayesian Stochastic Volatility (SV)

The second model is a Bayesian stochastic volatility (SV) model. This model treats volatility as a latent stochastic process. In the GARCH model volatility evolves deterministically based on past returns but the SV model allows volatility to evolve independently over time. providing greater flexibility in capturing persistent and smoothly varying changes in financial market uncertainty.
The Bayesian stochastic volatility model is defined as:

$$r_t = \exp\left(\frac{h_t}{2}\right)\varepsilon_t, \quad \varepsilon_t \sim t_\nu(0,1),$$
$$h_t = \mu + \phi(h_{t-1} - \mu) + \eta_t, \quad \eta_t \sim \mathcal{N}(0, \sigma^2).$$

The parameters to be estimated are:

- μ — long-run mean of log-volatility,
- ϕ — persistence of volatility,
- σ^2 — variance of the volatility innovations (volatility-of-volatility),
- ν — fixed degrees of freedom controlling tail heaviness.

In this implementation, ν is fixed (set to $\nu = 8$) to simplify inference while still allowing for heavy-tailed behavior in returns.

2.2.1 Parameter Estimation

Latent Parameter
SV has a latent volatility process $h_1, \ldots, h_n$, which is not observable. So, given the current parameter values $(\mu, \phi, \sigma^2, \nu)$, the h_i are sampled and updated using a MCMC approach. The conditional posterior of h_t depends only on:

- the observation r_t,
- the previous state h_{t-1},
- the next state h_{t+1}.

So, the conditional posterior is:

$$p(h_t \mid \cdot) \;\propto\; p(r_t \mid h_t)\, p(h_t \mid h_{t-1})\, p(h_{t+1} \mid h_t)$$

Where, Since:

$$\begin{aligned}
r_t \mid h_t \sim t_\nu\left(0, \exp(h_t)\right) \quad &\implies \quad p(r_t \mid h_t) = \frac{\Gamma\left(\frac{\nu+1}{2}\right)}{\Gamma\left(\frac{\nu}{2}\right)\sqrt{\nu\pi}\,\exp(h_t/2)}\left(1 + \frac{r_t^2}{\nu\,\exp(h_t)}\right)^{-(\nu+1)/2} \\
h_t \mid h_{t-1} \sim \mathcal{N}\big(\mu + \phi(h_{t-1} - \mu), \sigma^2\big) \quad &\implies \quad p(h_t \mid h_{t-1}) = \frac{1}{\sqrt{2\pi\sigma^2}}\exp\left(-\frac{\left[h_t - \mu - \phi(h_{t-1} - \mu)\right]^2}{2\sigma^2}\right) \\
h_{t+1} \mid h_t \sim \mathcal{N}\big(\mu + \phi(h_t - \mu), \sigma^2\big), \quad &\implies \quad p(h_{t+1} \mid h_t) = \frac{1}{\sqrt{2\pi\sigma^2}}\exp\left(-\frac{\left[h_{t+1} - \mu - \phi(h_t - \mu)\right]^2}{2\sigma^2}\right)
\end{aligned}$$

At each MCMC iteration, indexed by $m = 1, 2, \ldots, M$, and for each $t = 1, \ldots, n$, a new value for h_t is proposed according to:

$$h_t^* \sim \mathcal{N}(h_t^{(m-1)}, \tau^2),$$

where τ^2 is a tuning parameter that is adjusted based on acceptance ratio. Then the log of the posterior density at the current value ℓ_{curr} is calculated and compared to the log of the posterior at the proposed value ℓ_{prop}. The proposal is accepted with probability:

$$A = \min\left(1, \exp(\ell_{\text{prop}} - \ell_{\text{curr}})\right).$$

If accepted, set $h_t = h_t^*$, otherwise keep the previous value. A single MCMC iteration does a full sweep through $t = 1, \ldots, n$, updating each h_t sequentially.
The h_i are initialized using the observed returns. We set:

$$h_t^{(0)} = \log(r_t^2 + \epsilon),$$

where ϵ is a small constant added for numerical stability. (10^{-6} in our case).

Observed Parameters
We once again use the Metropolis–Hastings algorithm and follow the same adaptive proposal density construction approach proposed in [1] as used in 2.1.1. Where we again we map the constrained parameter space to an unconstrained space:

$$u = (u_1, u_2, u_3) \in \mathbb{R}^3$$

where:

$$\begin{aligned}
u_1 &= \mu \\
u_2 &= \log\left(\frac{\phi}{1-\phi}\right), \quad \phi = \frac{1}{1 + e^{-u_2}} \implies \phi \in (0,1) \\
u_3 &= \log\sigma^2 \quad \implies \sigma^2 = e^{u_3} > 0
\end{aligned}$$

Given the $h_{1:n}$, the parameters (μ, ϕ, σ^2) are sampled using a Metropolis-Hastings step. Conditioned on $h_{1:n}$:

$$h_t \mid h_{t-1} \sim \mathcal{N}\left(\mu + \phi(h_{t-1} - \mu), \sigma^2\right).$$

Thus, the log-likelihood is given by:

$$\log\mathcal{L} = -\frac{1}{2}\sum_{t=2}^{n}\left(\log(2\pi\sigma^2) + \frac{(h_t - \mu - \phi(h_{t-1} - \mu))^2}{\sigma^2}\right).$$

To ensure stationary of the process, we require:

$$|\phi| < 1, \quad \sigma^2 > 0$$

We place weakly informative priors:

$$\mu \sim \mathcal{N}(0, 10)$$
$$\phi \sim \text{Beta}(20, 1.5)$$
$$\sigma^2 \sim \text{Inverse-Gamma}(2, 0.1)$$

Assuming independence between parameters the full log posterior is:

$$\log p(\mu, \phi, \sigma^2, h_{1:n} \mid r_{1:n}) = \log \mathcal{L} + \log p(h_{1:n} \mid \mu, \phi, \sigma^2) + \log p(\mu, \phi, \sigma^2) + \log \left| \frac{\partial(\mu, \phi, \sigma^2)}{\partial(u_1, u_2, u_3)} \right|$$

Where:

$$\log p(h_{1:n} \mid \mu, \phi, \sigma^2) = \log p(h_1) - \frac{1}{2} \sum_{t=2}^{n} \left(\log(2\pi\sigma^2) + \frac{(h_t - \mu - \phi(h_{t-1} - \mu))^2}{\sigma^2} \right).$$
$$\log p(\mu, \phi, \sigma^2) = \log p(\mu) + \log p(\phi) + \log p(\sigma^2)$$
$$\left| \frac{\partial(\mu, \phi, \sigma^2)}{\partial(u_1, u_2, u_3)} \right| = \sigma^2 \cdot \phi(1 - \phi)$$

2.2.2 Algorithm

Unlike the GARCH model, the presence of the latent $h_{1:n}$ needs alternating updates between the $h_{1:n}$ and the model parameters. At each MCMC iteration the algorithm does the following in two steps:

1. **Latent variables update:** The $h_{1:n}$ are for each $t = 1, \ldots, n$, using the proposal and the posterior distribution derived in 2.2.1.
2. **Parameter update:** Given the updated $h_{1:n}$, the parameters (μ, ϕ, σ^2) are updated using the proposal and the posterior distribution derived in 2.2.1.

2.2.3 Results

The results of the simulation on our S&P 500 data are as follows:

Table 2: Bayesian Stochastic Volatility Model Parameters

Parameter	Mean	Median	2.5%	97.5%
μ	-10.5950	-10.5944	-10.5991	-10.5930
ϕ	0.2151	0.2153	0.2141	0.2156
σ^2	3.5289	3.5200	3.5130	3.5699

The posterior mean of $\mu \approx -10.60$ is the long-run average level of log-volatility. Since volatility is given by $\exp(h_t/2)$, this means a low daily volatility level. The posterior mean of $\phi \approx 0.22$ means moderate persistence in the volatility meaning that shocks decay fairly quickly over time. The posterior mean of $\sigma^2 \approx 3.53$ shows high variability in the volatility , meaning that volatility can change sharply from one period to the next.

The following is a graph of 5 simulated volatility paths against the real volatility path of our data:

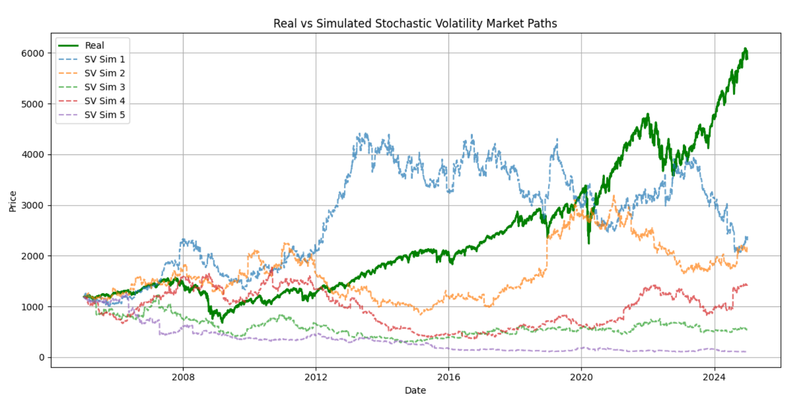


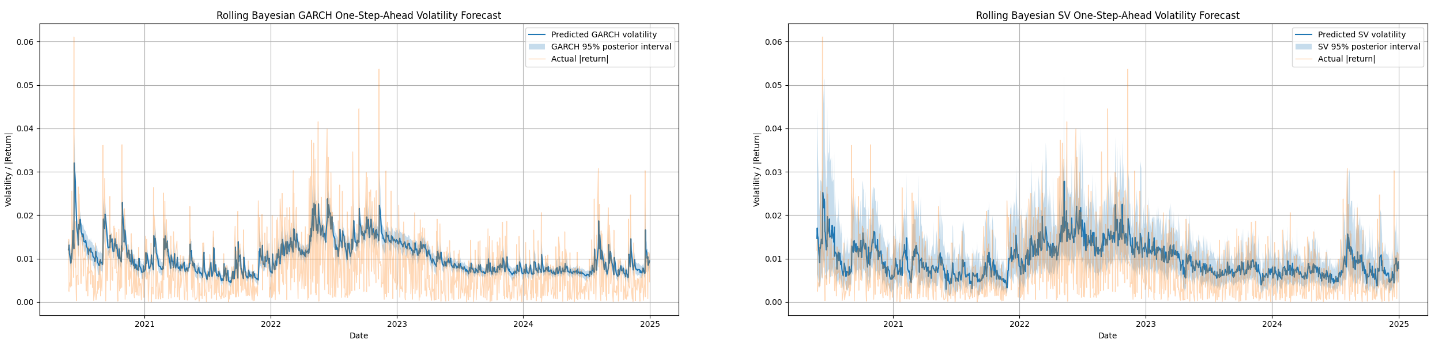
Figure 3: Real vs Simulated SV volatility paths

2.3 Model Comparisons

We compare the performance of our model in two ways, financial market implications and MCMC statistics.

2.3.1 Market Predictions

We run simulation on the last 5 years of our S&P 500 data, using rolling windows of size 100 to do parameter inference and predict volatility in the next time step. The results are as followed:



(a) GARCH Model

(b) SV Model

Figure 4: Rolling window prediction comparison for GARCH and SV models

GARCH produces smoother results with smaller confidence intervals around the prediction curve, Whereas SV produces more reactive results with bigger confidence intervals. This could mean that GARCH is more reliable to use in low volatility markets with smoother curves and SV is better to use in high volatility and crisis markets.

The SV model, particularly with Student-t innovations, is better suited for capturing heavy-tailed behavior in returns, leading to more realistic tail risk estimates compared to the Gaussian-based GARCH model. This improvement is important for risk management, where accurate estimation of Value-at-Risk (VaR) and Expected Shortfall (ES) depends on tail behavior.

2.3.2 Trading Strategy and Returns

We look at two different trading strategies based on the results of our models and compare them with Buy-And-Hold strategy.

1) Volatility and Portfolio Exposure

- if volatility is high, reduce total portfolio exposure
- if volatility is low, increase total portfolio exposure

Do this based on the inverse proportion of the market volatility and continuous adjustments. The results are as followed:

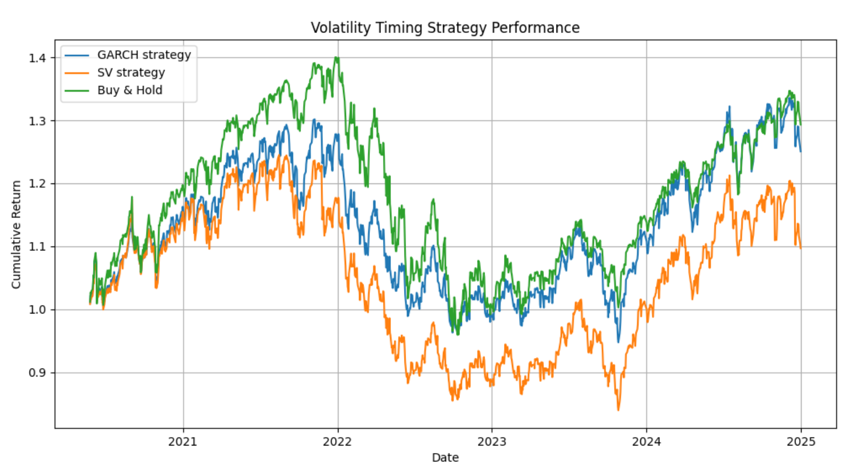


Figure 5: Volatility based Portfolio exposure strategy

Both SV and GARCH method under perform the S&P 500 over the long run, however the GARCH model does better comparative to the SV model and even peaks above the Buy-And-Hold curve slightly.

2) Binary Investment Regime Change

- if volatility is high, go cash
- if volatility is low, invest

The main difference between this strategy and the strategy above is binary adjustments with respect to a threshold. The results are as followed:

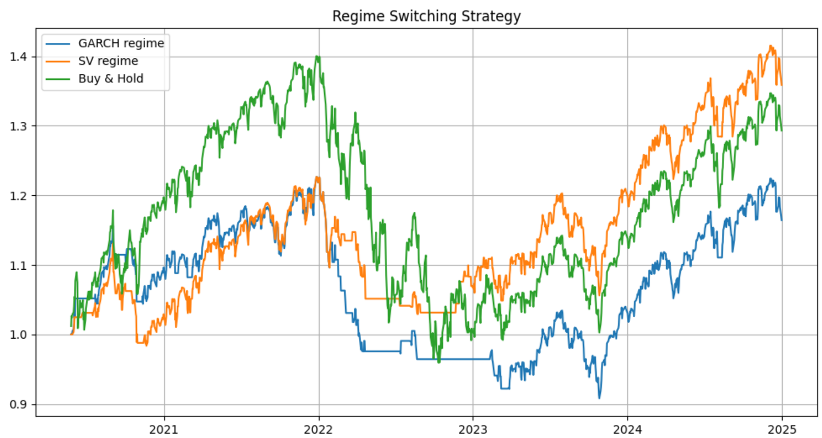


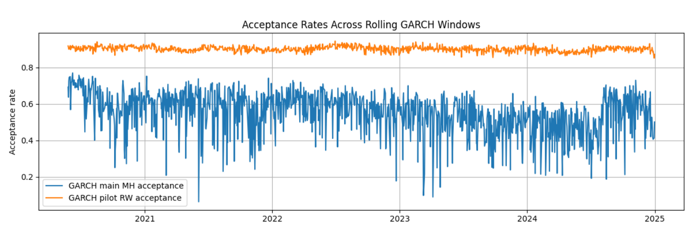
Figure 6: Volatility based Regime Change strategy

Initially Both SV and GARCH method under perform the S&P 500 , however in the long-run the SV outperforms the Buy-And-Hold strategy by around 7%. The GARCH model seems to make a bad decision early on and spends the rest of the time recovering from that mistake in which it does a good job and becomes eventually consistent with the market.

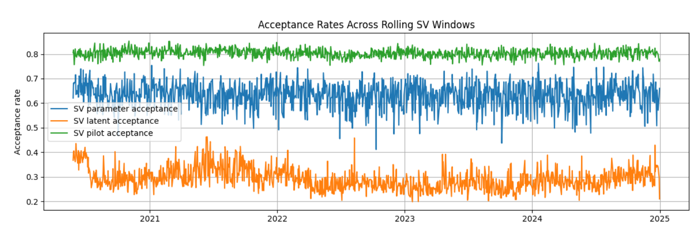
2.3.3 MCMC Performance

We evaluate the efficiency of the MCMC algorithms for both models using acceptance rates and general runtime.

1) Acceptance Rates
We evaluate the performance of the MCMC algorithms using acceptance rates across rolling windows. The following graphs outline the acceptance rates of our models across different phases.



(a) GARCH



(b) SV

Figure 7: Acceptance rates for GARCH and SV models over the full dataset

For the GARCH model, the pilot achieves pretty high acceptance rates (around 0.85-0.90), this means that the proposal covariance is good for initializing the adaptive method. The main sampling section has more variability, with acceptance rates anywhere between 0.4 and 0.7, occasionally dropping lower during periods of higher volatility. This can mean that while the adaptive proposal leads to exploration, the posterior geometry changes across rolling windows, making uniform tuning difficult.

For the stochastic volatility (SV) model, we see a more stable behavior across different phases. The pilot phase remains stable at approximately 0.8. The parameter updates acceptance rate is around 0.6-0.7, meaning there is efficient mixing. However, the latent state updates have much lower acceptance rates (around 0.25-0.35), which can be due to the high dimensionality of the latent volatility process. Lower acceptance rates in the latent state updates of the SV model also indicate slower mixing of the Markov chain, reflecting the increased difficulty of exploring a high-dimensional and strongly dependent posterior distribution.

2) Runtime
The GARCH model is computationally efficient, since volatility is computed recursively and only a small number of parameters are sampled. This results in low per-iteration cost across rolling windows.

In contrast, SV is significantly more computationally costly, because on top of sampling model parameters, the algorithm samples the whole latent volatility path for each iteration. This increases the dimension of the sampling problem and therefor the cost at each iteration, leading to longer runtimes.

3 Future Work

Among the many things that can be worked on for future iterations of this project these are the things that standout the most:

- Doing more rolling window analysis over low volitant markets, or other high volatility markets such at the 2009 financial crisis markets to see how each method performs.
- Analyze acceptance rate behavior by finetuning the proposal distribution by scaling the covariance matrix.
- using formal tail risk analysis, such as Value-at-Risk and Expected Shortfall tests.

4 Conclusion

In this project, we compared Bayesian GARCH(1,1) and stochastic volatility (SV) models for financial return data using MCMC methods for estimating model parameters. The results show that while the GARCH model provides computational efficiency and stable estimation, the SV model offers greater flexibility in capturing volatility at a cost of higher computational complexity.

The SV model produces better estimates of extreme market movements. This is important for risk management applications and accurately modeling rare but severe events.

Overall, the results show a trade-off between computational cost and modeling capacity. GARCH is good for fast and stable estimation, while SV provides a better framework for capturing volatility at a higher computational cost.

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